PRJ-001

Front Brake Rotor Thermal Optimisation, Project Report (G0–G5)

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Notice. Independent portfolio project, not affiliated with any vehicle or brake manufacturer. The rotor geometry is representative of the vehicle class, not an OEM drawing. Material data and boundary conditions are assumed or taken from open literature as stated. All results are simulation-only and have not been validated by physical test. Not for production use.

Revision history

Rev	Date	Description
A	2026-11-01	Initial release

Contents

1. Summary . . . 3
2. Concept: how a vented rotor sheds heat . . . 3
3. Programme . . . 5
4. Benchmark and reference vehicle (G1) . . . 6
5. Requirements and load cases (G2) . . . 6
6. Baseline design (G3) . . . 8
7. Concept phase (G3) . . . 9
8. Detailed thermal analysis (G4) . . . 10
   8.1 3-D FEA set-up . . . 10
   8.2 Verification . . . 10
   8.3 Lumped model vs 3-D . . . 10
   8.4 Uncertainty budget . . . 11
9. Vent airflow CFD (G4b) . . . 12
10. Two-zone thermal model (G5) . . . 12
11. Design of experiments and optimisation (G5) . . . 14
   11.1 Design variables and constraints . . . 14
   11.2 CFD design of experiments for the airflow . . . 14
   11.3 Search . . . 15
   11.4 Candidate selection and verification . . . 15
   11.5 The frozen design, G5C . . . 17
   11.6 Splitter vanes (post-freeze check) . . . 19
12. Manufacturability and the practical design . . . 20
   12.1 Why this matters . . . 20
   12.2 What was found . . . 20
   12.3 Trade-off . . . 20
   12.4 G5D against G5C . . . 20
13. What the result does and does not show . . . 21
14. AI-assisted working method: assessment . . . 22
15. Conclusions and recommendation . . . 22
16. Open items and next steps (G6) . . . 23
Appendix A. Document register . . . 23
Appendix B. Principal assumptions . . . 24

Reading guide. This report covers the project from charter to design freeze, in the order the work was done. Each gate section states what was done, what was checked and the result. Decisions are in boxes headed **DECISION**; engineering judgements that qualify a result are in boxes headed **ASSESSMENT**. Detailed methods and data are in the supporting documents listed in Appendix A, cited by document number.

1. Summary

Objective. Reduce the peak surface temperature of a ventilated front brake rotor for a C/D-segment SUV under repeated braking, without increasing rotor mass, inside the original envelope. Simulation only; public data and open-source tools only.

Result.

Design	Description	Peak surface temperature, LC-04 15 stops (3-D FEA)	Rise vs baseline	Mass (CAD)	Narrowest vent channel
C0 baseline	36 curved vanes × 4.5 mm, plates 9.0 mm	592.4 °C	reference	9.19 kg	9.5 mm
G5C (frozen, D-04)	62 straight radial vanes × 3.3 mm, plates 7.1 mm	**545.9 °C**	**–8.2 %**	**8.27 kg**	6.2 mm
G5D (thicker sections)	48 near-straight vanes × 4.0 mm, plates 8.0 mm	568.5 °C	–4.2 %	8.89 kg	7.8 mm
G5A (thermal bound)	64 swept vanes × 3.0 mm, plates 7.2 mm	529.4 °C	–11.1 %	8.45 kg	3.2 mm

Rise = peak surface temperature minus the 25 °C start temperature. Cooling model: CFD-calibrated vent airflow, Limpert vane heat-transfer correlation with an effectiveness cap (Sections 8 and 11).

Status of requirements (G5C). All G5 requirements are met except REQ-THM-001, the 10 % stretch target (–8.2 % achieved). The NEDC margin (0.4 °C) is inside the model uncertainty. Manufacturability (minimum vane, plate and channel sections) is not verified: the limits used are assumptions (Section 12).

Scope of the result. G5C is the best design found for one objective (LC-04 peak temperature at no added mass) with this project's models. It is not a production recommendation: noise, thermal cracking, pumping loss, cost and castability were not optimised, and the CFD heat-transfer check is not in the thermal model (Section 13).

Decisions taken.

ID	Gate	Decision	Record
D-01	G1	Reference vehicle class and baseline rotor Ø320 × 28 mm	DR-001
D-02	G3	C1 lead concept, C5 alternate; 10 % reduction kept as a stretch target	DR-002
D-03	G4	Measure vent airflow with CFD (OpenFOAM) before optimising	DR-003
G4 / G4b	G4	3-D model and uncertainty budget accepted; CFD cooling adopted	gate plan
D-04	G5	Freeze G5C	DR-004
D-05	G5	Production-intent variant: G5C or G5D, subject to foundry review	**open**, DR-005 (Section 12.4)

2. Concept: how a vented rotor sheds heat

This section states the physical mechanism the design work acts on. The later sections quantify it.

Heat path. During a stop the pads convert the vehicle's kinetic energy into heat at the two friction faces. About 90 % of it goes into the disc (assumed partition). A stop lasts a few seconds, too short for much cooling, so the heat is first stored in the friction ring, and the temperature rise of a single stop is set by the ring's heat capacity (mass × specific heat). Between stops the stored heat leaves by four routes:

- convection to the air pumped through the vent channels;
- convection from the outer friction faces and the hat;

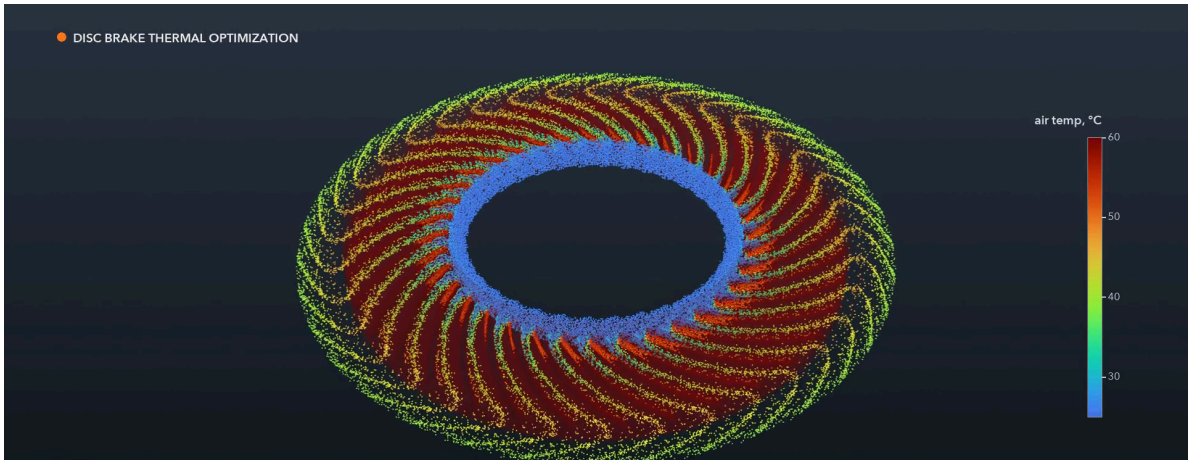


Figure 1: Baseline vent airflow coloured by air temperature, CFD at 80 km/h. Air enters cool at the centre and warms on its way to the rim. Illustration: rotor walls held at 400 °C, air at 25 °C (CFD-002)

- radiation from the exposed surfaces;
- conduction into the hat and the hub.

The envelope and the friction faces are fixed by the requirements. The internal geometry (vanes, vent gap, plate thickness) therefore acts mainly on the first route, and on how much heat the ring can store.

The rotor as a centrifugal pump. The vanes act like the blades of a centrifugal impeller. Rotation throws the air in the channels outward, so air is drawn in at the inner diameter of the vent and leaves at the rim. On the way it takes heat from the inner faces of the two plates and from the vane flanks. Figure 1 shows this for the baseline at 80 km/h; Figure 2 follows the air through one channel.

Governing relations. The channel walls hand heat to the air at the rate

$$\dot{Q}_{\text{vent}} = h\, A_{\text{wet}}\, (T_{\text{wall}} - T_{\text{air}}) \tag{1}$$

and the air carries it away with a temperature rise

$$\Delta T_{\text{air}} = \frac{\dot{Q}_{\text{vent}}}{\dot{m}\, c_p} \tag{2}$$

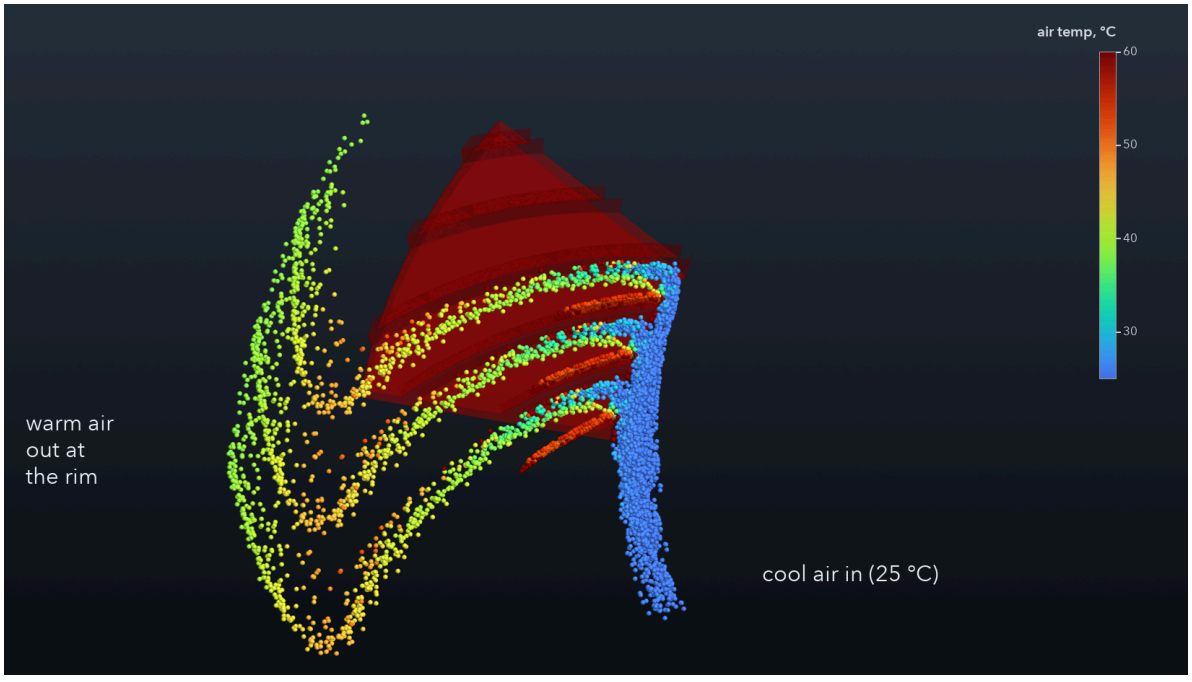


Figure 2: One vent channel of the baseline, close up (same CFD case). Air enters at ambient temperature (blue) and picks up heat from the plates and vanes as it flows outward. Mean air temperature at r = 105 / 125 / 145 mm: 49 / 55 / 57 °C

Here h is the heat-transfer coefficient, A_{wet} the wetted area of the channels, $\dot{m}$ the mass flow of air through the vents and c_p the specific heat of air. The two relations are coupled: as the air warms along the channel, the driving difference $T_{\text{wall}} - T_{\text{air}}$ falls. Combined over the channel length, they give the heat-exchanger form used in the thermal model (Section 11.3):

$$\dot{Q}_{\text{vent}} = \varepsilon\, \dot{m}\, c_p\, (T_{\text{wall}} - T_{\text{in}}), \qquad \varepsilon = 1 - e^{-\text{NTU}}, \qquad \text{NTU} = \frac{h\, A_{\text{wet}}}{\dot{m}\, c_p} \tag{3}$$

The effectiveness ε cannot exceed 1. Once the air leaves nearly as hot as the walls ($\text{NTU} \gg 1$), more area or a higher h removes little extra heat, and only more air flow helps.

The heat stored in one stop sets the temperature rise of that stop:

$$\Delta T_{\text{stop}} \approx \frac{E_{\text{stop}}}{m\, c} \tag{4}$$

where E_{stop} is the braking energy into the disc, m the mass of the friction ring and c the specific heat of the iron.

Quantity	Physical meaning	Increased by	Penalty or limit
$\dot{m}$, vent air mass flow	How much air the rotor pumps	Radial rather than swept vanes, taller vent gap, more vanes up to a point	Narrower channels; pumping loss
h, heat-transfer coefficient	How readily the walls give heat to the air	Faster air in the channels	Follows $\dot{m}$ and channel size
A_{wet}, wetted area	Plate inner faces and vane flanks in contact with the air	More and longer vanes	Narrower channels (castability); less flow per channel
$m\, c$, heat capacity of the ring	Heat a stop can deposit for a given temperature rise	Thicker plates	Mass limit (no added mass)

The design problem. The outside size and the mass are fixed. Metal moved from the plates into more or thinner vanes increases $\dot{m}$ and A_{wet}, which speeds up cooling between stops. It also reduces heat capacity, so each stop heats the ring more. Which effect wins depends on the duty: over a long sequence of stops cooling dominates, while for one or two stops heat capacity dominates. This is why the optimised designs run hotter than the baseline in the first stops of LC-04 and cooler from then on (Section 11.5). A further limit comes from casting: thinner vanes and plates and narrower channels are harder to cast (Section 12).

> **ASSESSMENT.** Figures 1 and 2 illustrate the mechanism. They come from a CFD case with an assumed boundary condition (all rotor walls held at 400 °C), not a coupled thermal solution. The heat the air picks up in this CFD is 2–3 times below the correlation the thermal model uses, and the cause is not resolved (Section 9). The CFD heat-transfer magnitude is therefore **not used** in the thermal model. The CFD air flow ($\dot{m}$, as φ) is used.

3. Programme

Approach. The project was run as a stage-gated programme (PRG-001, PRG-002): G0 tools, G1 benchmark, G2 requirements, G3 concept, G4 detailed CAE, G4b airflow CFD, G5 optimisation and freeze, G6 virtual validation. Each gate has deliverables and an owner decision. Requirements trace to a verification plan (DVPR-001).

Tools (all open source, qualified at G0, PRG-003): CadQuery (CAD), Gmsh (meshing), CalculiX (3-D thermal FEA), OpenFOAM v2606 (CFD), SciPy (lumped thermal models, duty cycles), Optuna (optimisation), PyVista and Matplotlib (post-processing), Pandoc and Typst (reports).

Method of work. One engineer with an AI coding assistant (Claude Code). The assistant wrote most scripts and drafts; every result was cross-checked by an independent route (second model, energy balance, mesh study, literature, CFD). Errors found in AI-produced work are recorded in LOG.md (22 entries to G5, of which 2 record gate decisions; Section 14).

Limits that apply throughout.

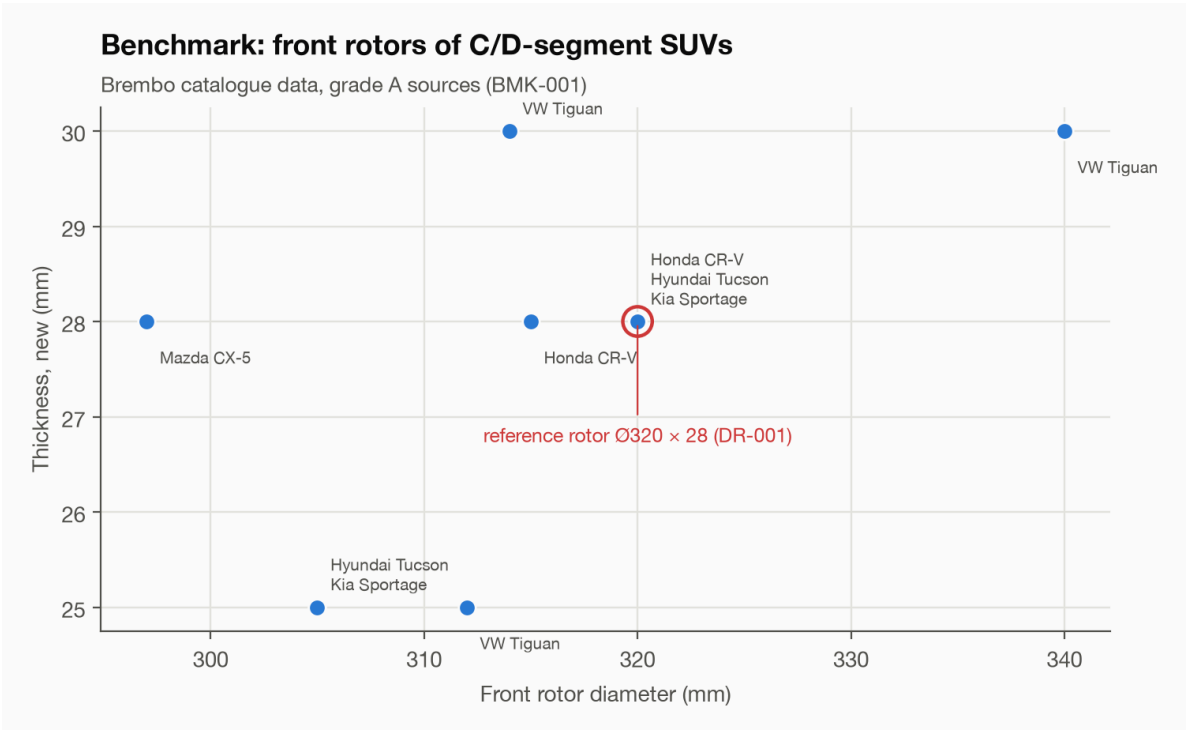


Figure 3: Front rotor sizes of six C/D-segment SUVs (BMK-001)

- Geometry is representative of the vehicle class, not an OEM drawing (D-01).
- Material properties are temperature-dependent grey-iron values from secondary sources (TM-001).
- Heat partition to the disc (90 %), emissivity (0.6) and front brake bias (70 %) are assumed.
- No test data exist. All verification is model-to-model or model-to-literature.

4. Benchmark and reference vehicle (G1)

Front rotors of six C/D-segment SUVs were taken from a manufacturer's parts catalogue (BMK-001). All are ventilated. Diameters range from 297 to 340 mm and cluster at 305–320 mm; thickness is 25 mm (base) or 28–30 mm (larger option). The pair Ø320 × 28 mm occurs on three of the six vehicles. The wear limit is typically 2 mm below new thickness.

Vehicle data (VTS-001): gross vehicle mass 2,150 kg for severe cases (one public source), test mass 1,850 kg for drive cycles (assumed), front brake bias 70 % (assumed), tyre rolling radius 0.35 m (assumed). A 100–0 km/h stop at 2,150 kg puts about 290 kJ into each front disc before the pad/disc heat split.

> **DECISION D-01 (G1, 2026-10-27, DR-001).** Reference rotor Ø320 × 28 mm, new; 26 mm wear limit. Vehicle masses as above.
>
> **Basis:** most common size in the benchmark for the upper trim of the class.

5. Requirements and load cases (G2)

Load cases (LC-001):

ID	Case	Purpose
LC-01	WLTC class 3b, test mass	Regulatory drive cycle
LC-02	NEDC, test mass	Legacy drive cycle
LC-03	WLTC × 3, no cool-down	Heat accumulation

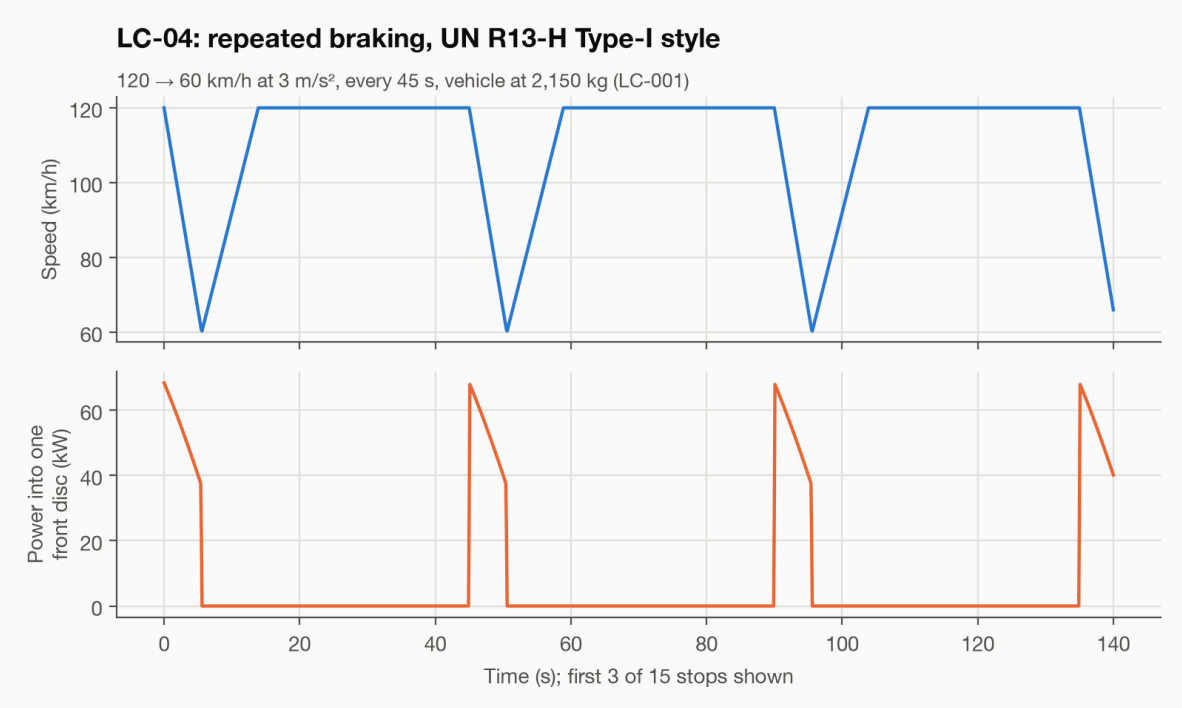


Figure 4: LC-04 speed and power into one front disc, first three of fifteen stops

ID	Case	Purpose
LC-04	15 stops 120 → 60 km/h at 3 m/s², every 45 s, GVM (after the UN R13-H Type-I heating procedure)	**Primary design case** (fade)
LC-05	Descent: 50 km/h, 10 % grade, 6 km, GVM, friction brakes only	Sustained heating (project-defined)
LC-06	Single stop 100 → 0 km/h at 1 g, GVM	Peak power; model calibration
LC-07	10 stops 100 → 0 km/h at 0.6 g, every 30 s, GVM	Repeated hard stops (project-defined)

Requirements (SRS-001, abbreviated):

ID	Requirement	Basis
REQ-GEO-001	Fit the baseline envelope: OD ≤ 320 mm, thickness ≤ 28 mm, same hat and bolt interface	Drop-in replacement
REQ-GEO-002	Mass ≤ baseline (CAD)	Unsprung mass
REQ-GEO-003	Thermal requirements also met at the 26 mm wear limit	Rotors run worn (G6)
REQ-THM-001	LC-04 peak temperature rise ≥ 10 % below baseline (stretch target, D-02)	Project goal
REQ-THM-002	LC-01/02/03 peak ≤ baseline	No regression on drive cycles
REQ-THM-003	LC-04/05/07 peak ≤ 700 °C	Literature: grey-iron microstructure change near 700 °C
REQ-THM-004	Convective cooling capability at 100 km/h ≥ baseline	Cooling, not only mass
REQ-MDL-001	Lumped model vs 3-D FEA within the validation uncertainty (ASME V&V 20 approach)	Model trust
REQ-MDL-002	Mesh uncertainty reported as GCI on ≥ 3 meshes, ≤ 2 %	Model trust

No standard sets a rotor temperature limit or an improvement target; the 700 °C limit is literature-based and the 10 % target is a project choice. Both are marked as such in SRS-001.

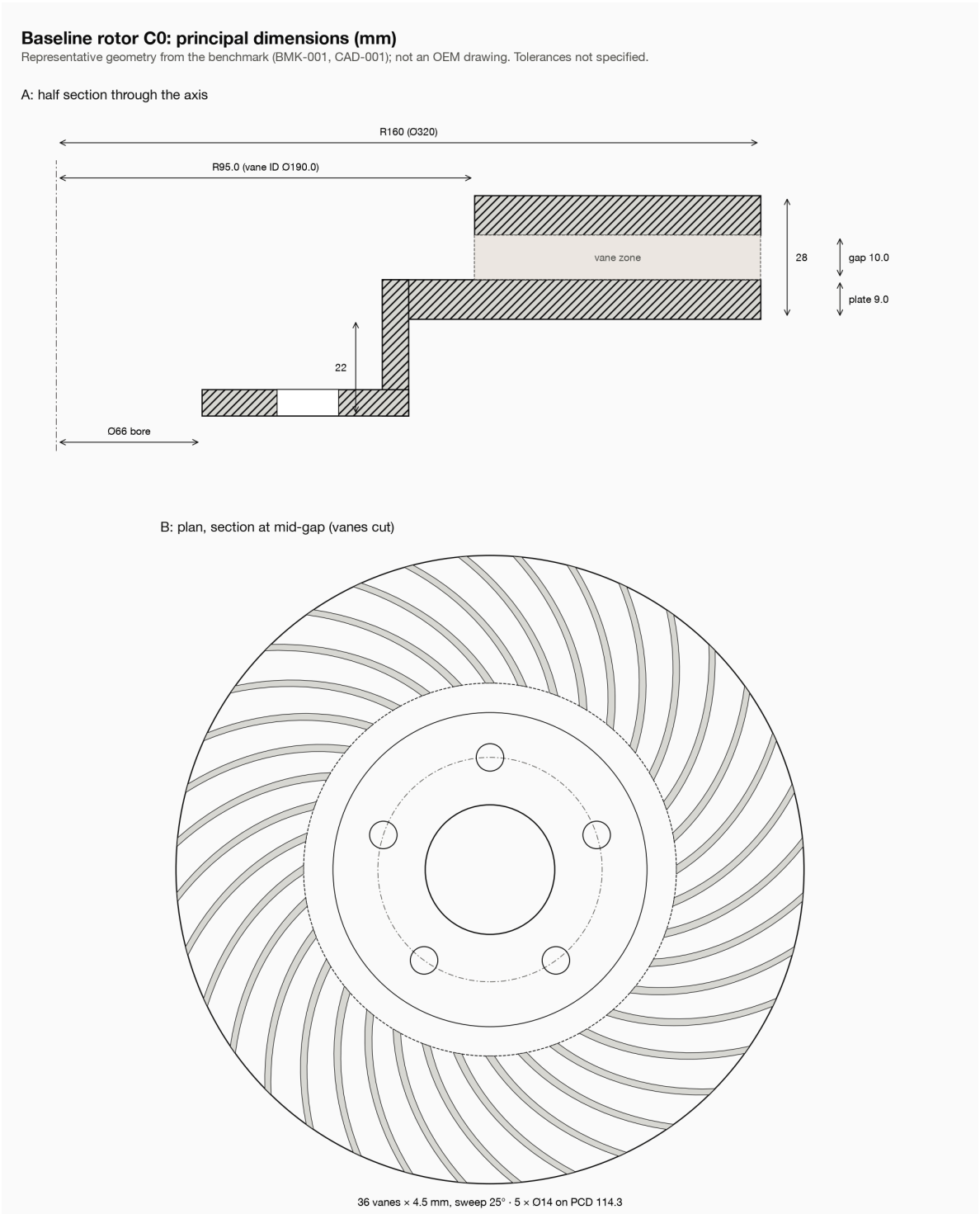


Figure 5: Baseline rotor C0, principal dimensions in mm (CAD-001)

6. Baseline design (G3)

The baseline was built as a fully parametric CadQuery model (CAD-001). Main dimensions are from the benchmark (D-01); vane count, thickness and sweep, plate thickness and vane inner diameter are representative values, marked ASSUMED.

Baseline: 36 curved vanes, 4.5 mm thick, swept 25° from inner to outer radius; friction plates 9.0 mm; vent gap 10.0 mm; vanes from Ø190 mm to the rim; CAD mass 9.19 kg (grey iron, 7,150 kg/m³).

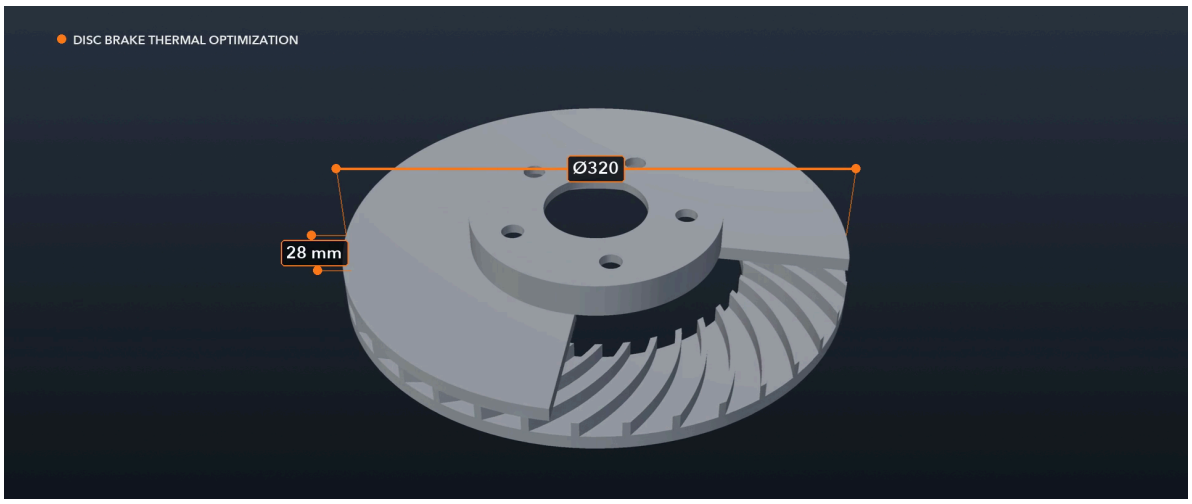


Figure 6: Baseline rotor, parametric CAD model (CadQuery)

7. Concept phase (G3)

Model. A lumped (1-D through-thickness) thermal model of the friction ring (TM-001): conduction through plate and vent zone, convection from the faces and vent channels, radiation, temperature-dependent material data. Vent air speed was set as $v = \varphi\,\omega\,r_{\text{out}}$ with an **assumed** flow coefficient $\varphi = 0.4$. Run time is seconds per load case.

Concepts (CON-001), LC-04 peak rise vs baseline, lumped model, G3 cooling assumptions:

Concept	Change from baseline	Mass change	LC-04 rise change
C1	48 vanes × 3.5 mm	+0.02 kg	–3.7 %
C2	Plates 8 mm, gap 12 mm	–0.68 kg	+0.4 %
C3	Straight radial vanes (0° sweep)	0.00 kg	+2.8 %
C4	24 vanes × 5 mm, 40° sweep	–0.20 kg	+1.4 %
C5	Plates 8 mm, gap 12 mm, 48 × 3.5 mm vanes	–0.64 kg	–4.2 %

DECISION D-02 (G3, 2026-10-27, DR-002). C1 lead concept, C5 alternate, to 3-D FEA. The 10 % reduction target is not reachable by these concepts; it is kept as a stretch target and the achieved value is reported. The design space is widened at G5.

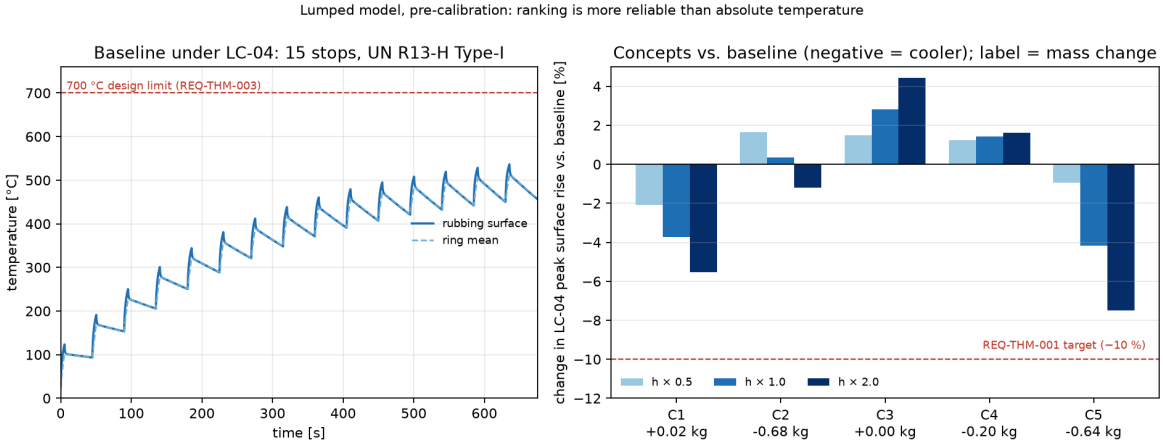


Figure 7: Concept comparison, LC-04 peak surface temperature, lumped model, G3 cooling (CON-001)

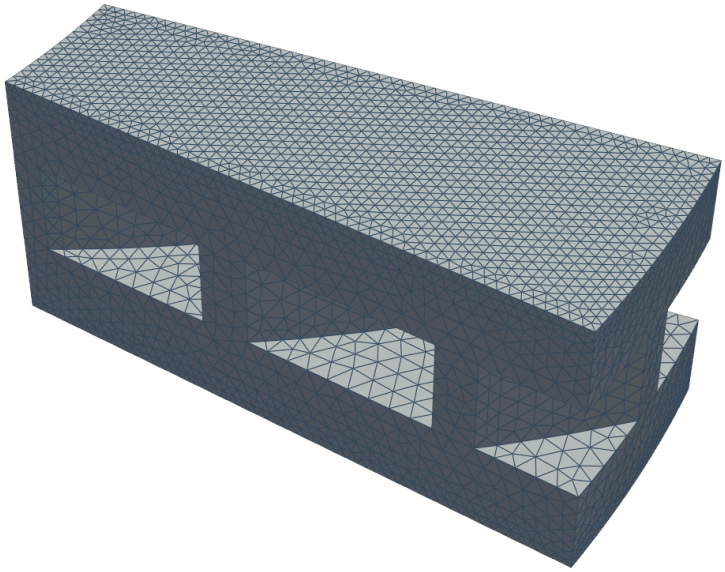


Figure 8: 3-D FEA: one 10° sector of the friction ring, coarse mesh (CAE-001)

ASSESSMENT (made at G5). C3, straight radial vanes, ranked worst here because the G3 model held ϕ constant: it could not credit straight vanes with more airflow. The CFD of G4b–G5 showed that sweep is the strongest negative factor on airflow (Section 11), and the frozen design G5C has straight radial vanes. A fixed-ϕ assumption steered the G3 ranking; this is recorded as a model-form error of the concept phase.

8. Detailed thermal analysis (G4)

8.1 3-D FEA set-up

CAE-001. One periodic sector of the friction ring (10° for 36 vanes) meshed with tetrahedra in Gmsh and solved transient in CalculiX. Friction heat is applied on the pad track (r = 100–150 mm), on both faces. Convection and radiation as in the lumped model. The hat is not modelled.

8.2 Verification

Check	Result
Mesh: 3 meshes (37k / 114k / 335k elements), LC-06	Peak changes 0.04 % between the two finest; GCI 0.11 % (REQ-MDL-002 met)
Time step: 0.1 s vs 0.05 s	Peak changes 0.15 %
Energy: heat in = stored + convected + radiated	Closed to < 0.5 % after a fix (below)
Mass: sector × 36 vs lumped model	Disagreement of 3 % exposed an error in the lumped vane area (fixed)

8.3 Lumped model vs 3-D

Quantity	Result
Ring mean temperature	Agrees to 0.0–0.2 %
Peak surface, heat spread over the ring (like for like)	Lumped low by 0.8–3.8 % of the rise (REQ-MDL-001 met)
Peak surface, heat on the pad track	3-D higher by 38 % (1 stop), 15 % (3 stops), 6.7 % (15 stops)
Concept ranking, 15 stops	Correct; C1 –3.5 %, C5 –3.6 % in 3-D

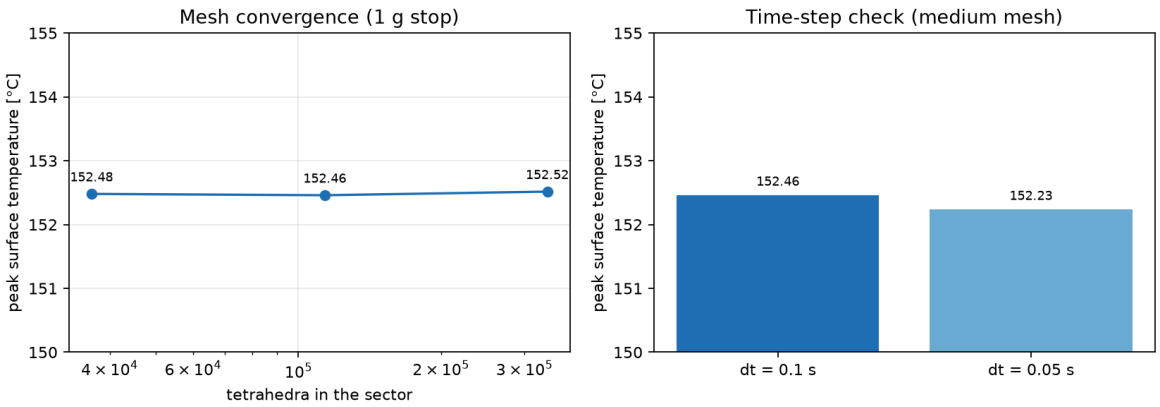


Figure 9: 3-D FEA mesh and time-step convergence (CAE-001)

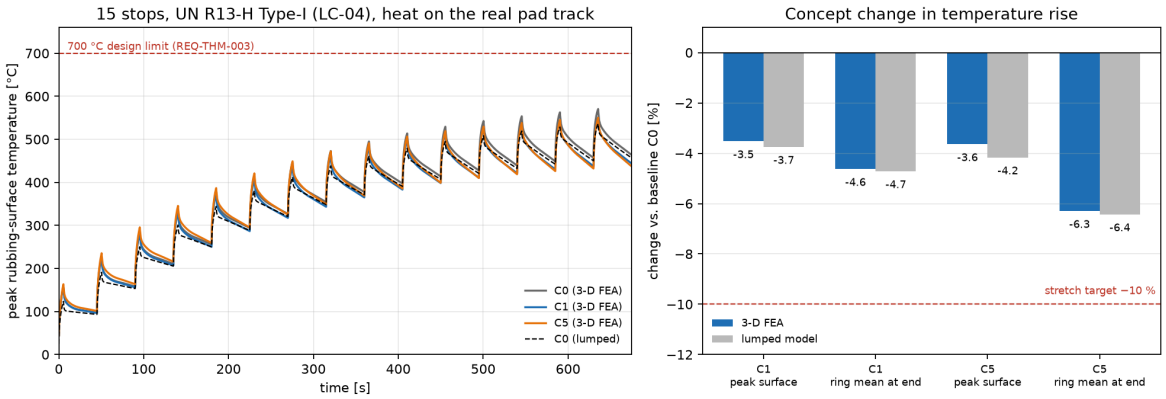


Figure 10: LC-04, 15 stops: 3-D FEA against the lumped model (CAE-001)

ASSESSMENT. The lumped model is adequate for ranking designs and for energy, but under-predicts the absolute surface peak because it spreads the heat over the whole ring. Absolute temperatures are taken from 3-D FEA.

8.4 Uncertainty budget

VV-001, ASME V&V 20 approach (model-to-model; there is no test data). Numerical uncertainty about 0.5 °C. Input uncertainty from a 256-sample Monte Carlo over nine inputs with assumed ranges: the cooling inputs (heat-transfer correlations and vent flow coefficient) accounted for 74 % of the variance of the baseline temperature rise, the vent flow coefficient alone 27 %.



Figure 11: Baseline temperature field during LC-04, 3-D FEA (sector replicated for display)

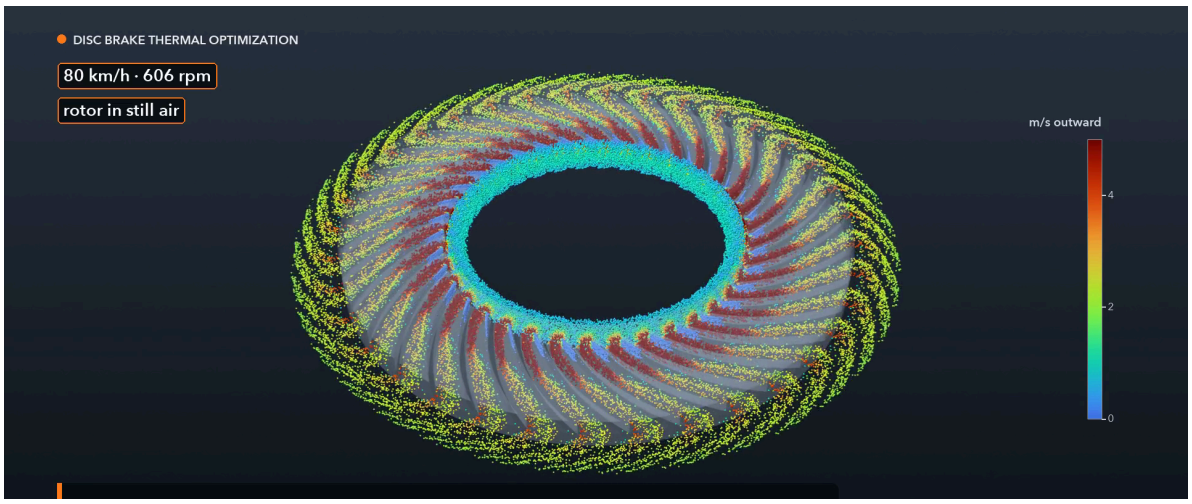


Figure 12: Vent airflow of the baseline at 80 km/h, CFD; particles coloured by outward air speed (CFD-001)

DECISION (G4, 2026-10-30, owner). G4 passed. 3-D model and uncertainty budget accepted.

DECISION D-03 (G4, DR-003). The vent flow coefficient, until then an assumption, is to be measured with CFD before optimisation.

9. Vent airflow CFD (G4b)

CFD-001. OpenFOAM, steady RANS, k-ω SST with wall functions, rotating reference frame, one periodic sector of the rotor in still air, 80 km/h (606 rpm). Output: the air speed through the vent channels, expressed as ϕ.

Check	Result
Mesh (wall cells 1.0 / 0.7 / 0.5 mm)	ϕ 0.218 / 0.238 / 0.247; extrapolated 0.254, GCI 3.7 %
Road speed 40 / 80 / 120 km/h	ϕ 0.231 / 0.238 / 0.241: independent of speed, as the model form assumes
Domain twice as large	–0.2 %
Mass balance through the open boundaries	1e-8 of the flow
Literature: Limpert, Brake Design and Safety (vane-flow equation)	17.9 g/s vs CFD 17.0 g/s (–5 %)

Effect. ϕ = 0.255 replaced the assumed 0.40. The vent heat-transfer correlation was changed to Limpert's laminar vane correlation (the channel Reynolds number is 900–2,700). The 3-D baseline peak in LC-04 rose from 570.5 to 584.4 °C; the concept ranking was unchanged (C1 –3.6 %, C5 –3.2 %). The uncertainty budget was repeated: ϕ fell from 27 % to 7 % of the variance; the heat-transfer correlation scale is now the largest term (55 %).

Open item from G4b. A passive-temperature CFD run (walls at 400 °C) gave 2–3 times less vent heat pick-up than the correlations; a finer mesh moved further away, not closer (CFD-002 Sec. 5). The cause is most likely the near-wall mesh (no prism layers). This CFD heat transfer is **not used** in the thermal model.

DECISION (G4b, 2026-10-30, owner). G4b passed. CFD airflow and the Limpert correlation adopted as the nominal cooling. The CFD heat-transfer discrepancy carried open.

10. Two-zone thermal model (G5)

TM-002. The lumped model was split into two columns, pad track and remainder, coupled by radial conduction, so that it reproduces the pad-track hot spot of Section 8.3. One coupling parameter was calibrated on short

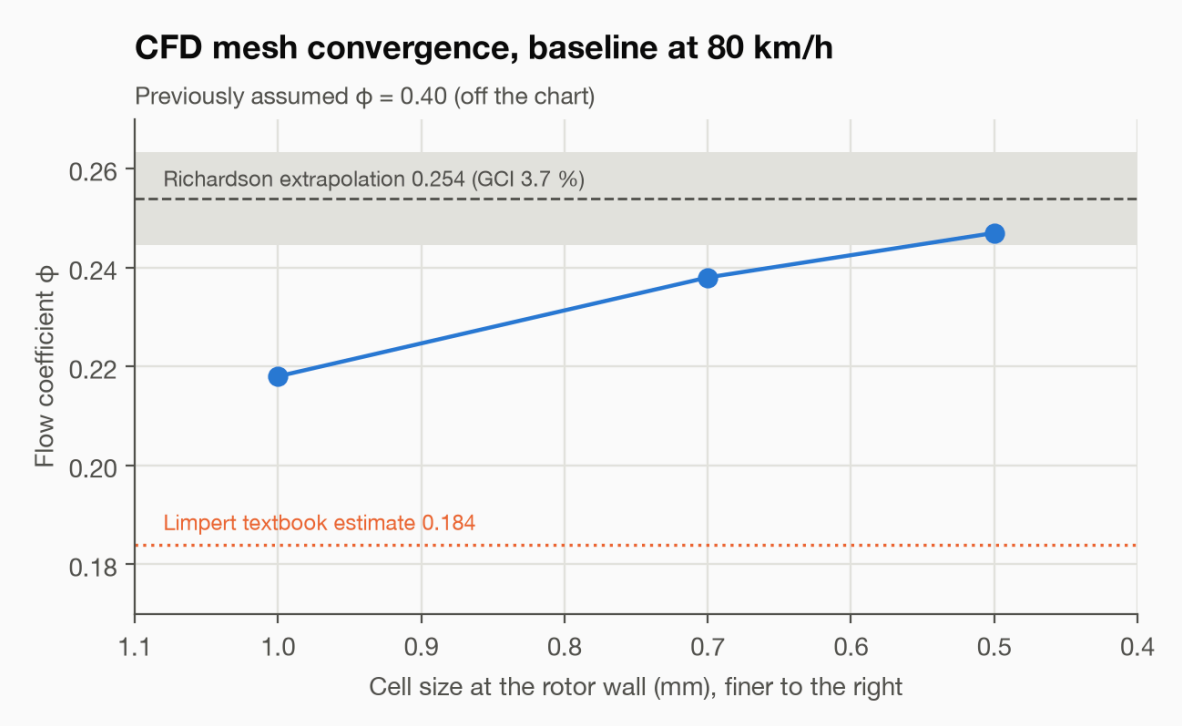


Figure 13: CFD mesh convergence of the flow coefficient ϕ, baseline at 80 km/h

cases (1 and 3 stops) and checked on held-out cases: the 15-stop peak within 0.0–0.1 % of the rise, and the same accuracy under different cooling levels. An effectiveness (NTU) limit was added to the vent heat transfer in G5 (Section 11.3). Run time is about 0.3 s per design for LC-04, which makes a search of thousands of designs practical.

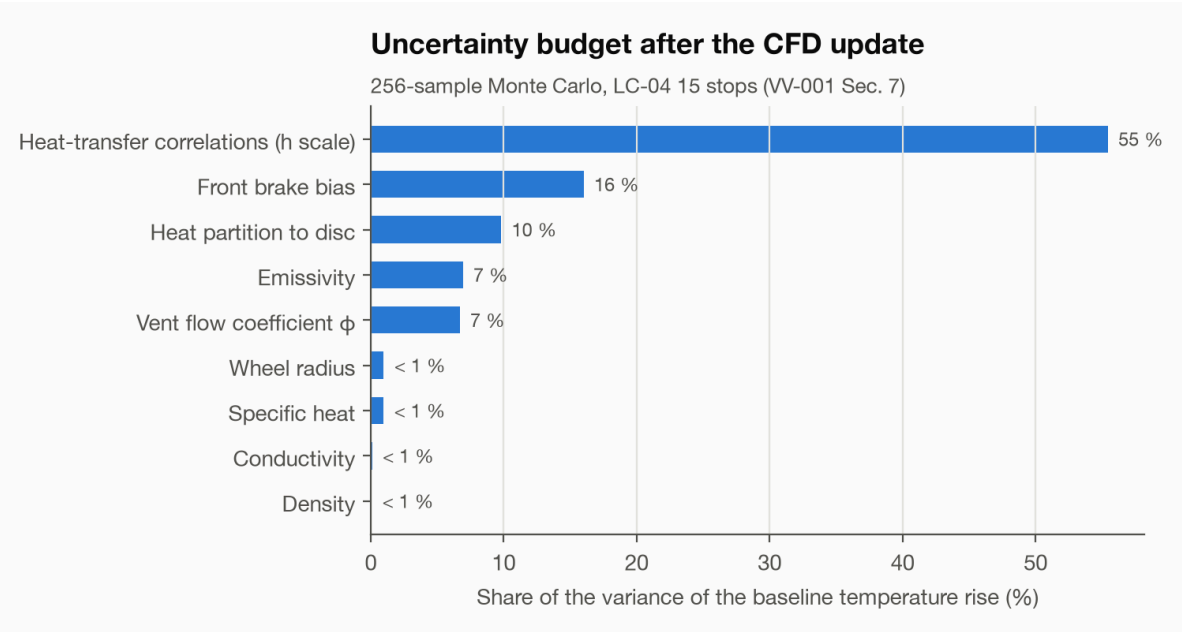


Figure 14: Variance shares of the baseline temperature rise after the CFD update (VV-001)

11. Design of experiments and optimisation (G5)

11.1 Design variables and constraints

Variable	Range	Baseline
Friction plate thickness (each)	7–11 mm	9.0 mm
Number of vanes	24–64	36
Vane thickness	3–6 mm	4.5 mm
Vane sweep, inner to outer radius	0–45°	25°
Vane inner diameter	175–205 mm	190 mm

Fixed: OD 320 mm, thickness 28 mm, hat, bolt pattern, pad track. Objectives: minimise LC-04 peak rise and mass. Constraints: mass ≤ baseline, cooling capability ≥ baseline, peak ≤ 700 °C; from round 4, narrowest vent channel ≥ 6 mm.

11.2 CFD design of experiments for the airflow

CFD-002. The search needs ϕ for every geometry. Sixteen designs spread evenly over the five variables (scrambled Sobol sequence), plus the baseline and concepts C1 and C5, were run with identical CFD settings (0.8 mm wall cells). ϕ ranged from 0.156 to 0.394, a factor of 2.5.

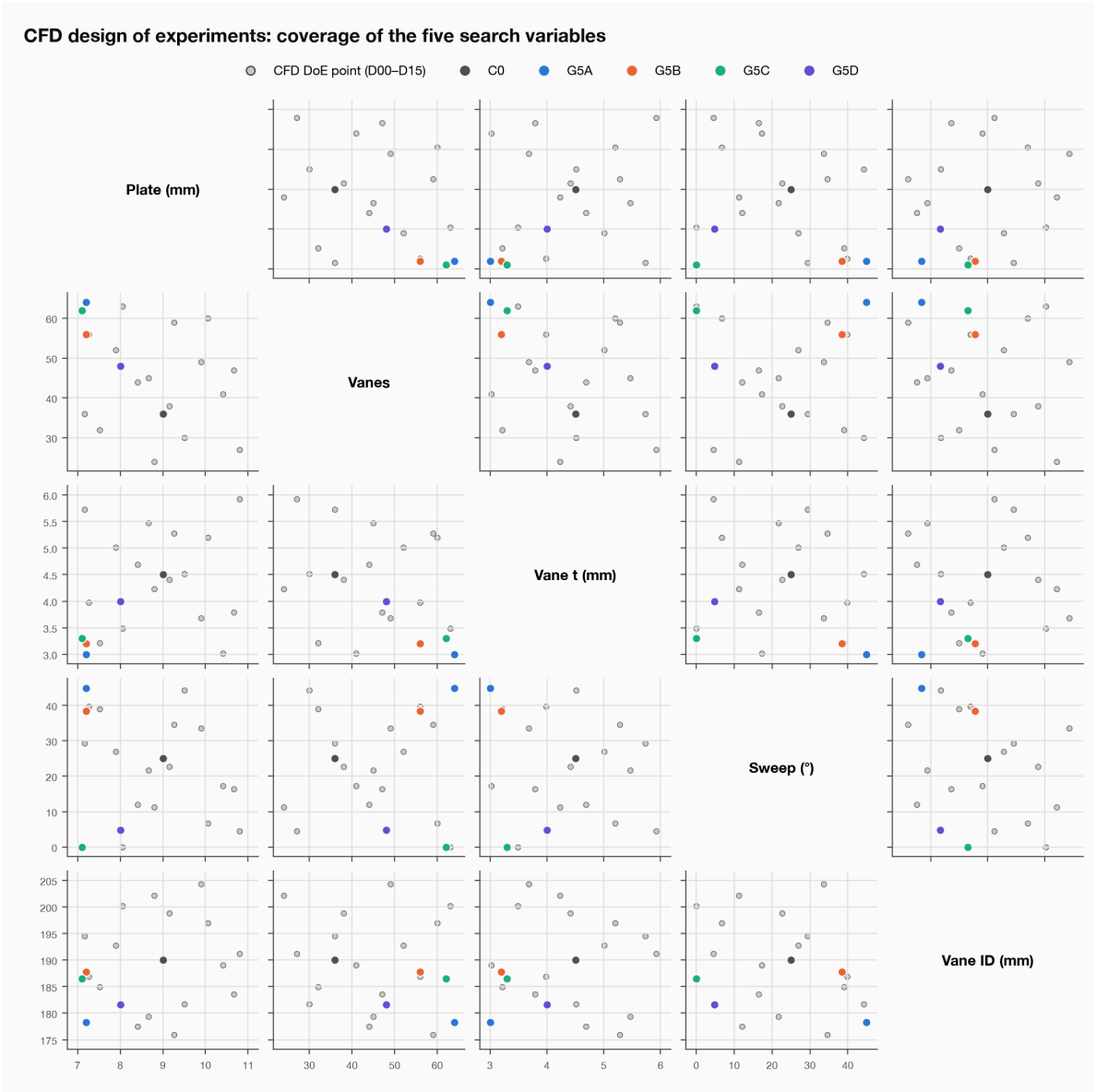


Figure 15: CFD design of experiments: pairwise coverage of the five design variables, with the candidates

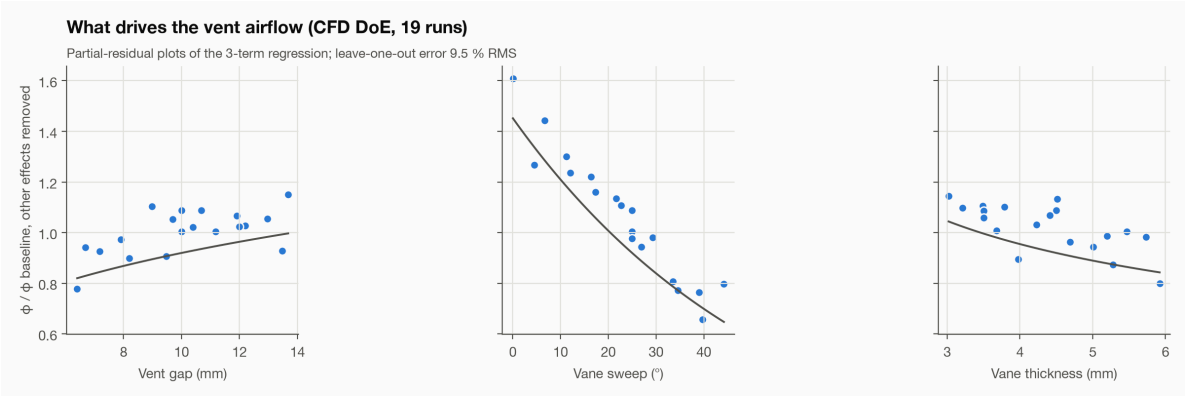


Figure 16: Effect of vent gap, vane sweep and vane thickness on ϕ (partial residuals of the regression)

A linear regression was chosen by leave-one-out error among seven candidate features:

$$\ln \frac{\varphi}{\varphi_0} = b_0 + b_1 \ln s_{\text{gap}} + b_2\, \theta_{\text{sweep}} + b_3 \ln t_{\text{vane}} \tag{5}$$

with vent gap s_{gap} (positive effect), vane sweep θ_{sweep} (strongest, negative) and vane thickness t_{vane} (negative); φ_0 is the baseline on the same mesh. Leave-one-out error 9.5 % RMS.

11.3 Search

NSGA-II genetic algorithm (Optuna), 28 designs per generation, about 900 designs per round, two-zone model with the regression ϕ. Four rounds were needed; in each, the optimum exploited a gap in the model that was closed before the next round.

Round	Model change	Predicted optimum	Gap the optimum exploited
1	ϕ from Limpert's estimate	−18.0 %	Textbook ϕ hardly falls when channels are nearly blocked
2	ϕ from the CFD regression	−15.8 %	Vent heat-transfer coefficient unbounded: more heat removed than the air flow can carry
3	Effectiveness (NTU) cap on vent heat transfer	−12.5 % (G5A)	Channels 3.2 mm wide; no castability limit in the model
4	Narrowest channel ≥ 6 mm	−8.1 % (G5C at −7.9 %)	none found; verified (Section 11.4)

Rounds 1–3 converged on the edge of the design space (64 vanes, 45° sweep, 7 mm plates). Round 4 converged on straight radial vanes, where the CFD data are dense.

11.4 Candidate selection and verification

Candidates were taken from the Pareto fronts: G5A and G5B (round 3), G5C (round 4) and G5D (DFM-conservative round, Section 12.3). Each received its own CFD run and a 3-D FEA run of LC-04, with identical settings for the baseline.

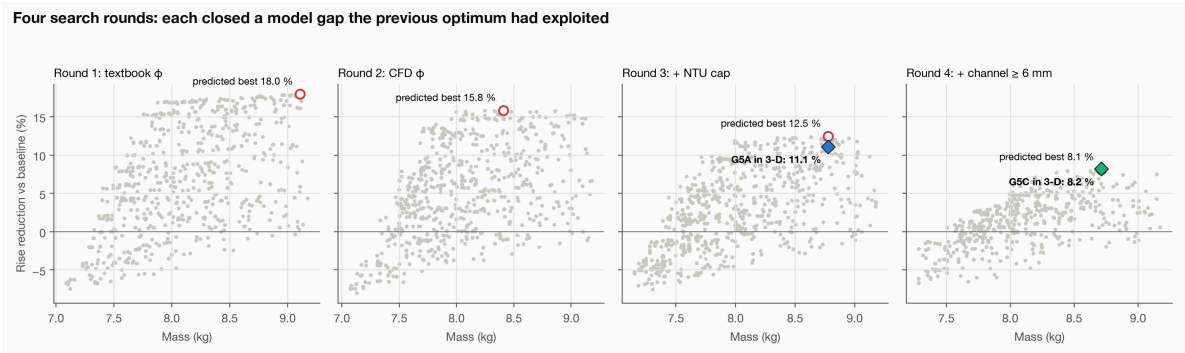


Figure 17: Feasible designs and predicted optimum of each search round, with the 3-D verified value

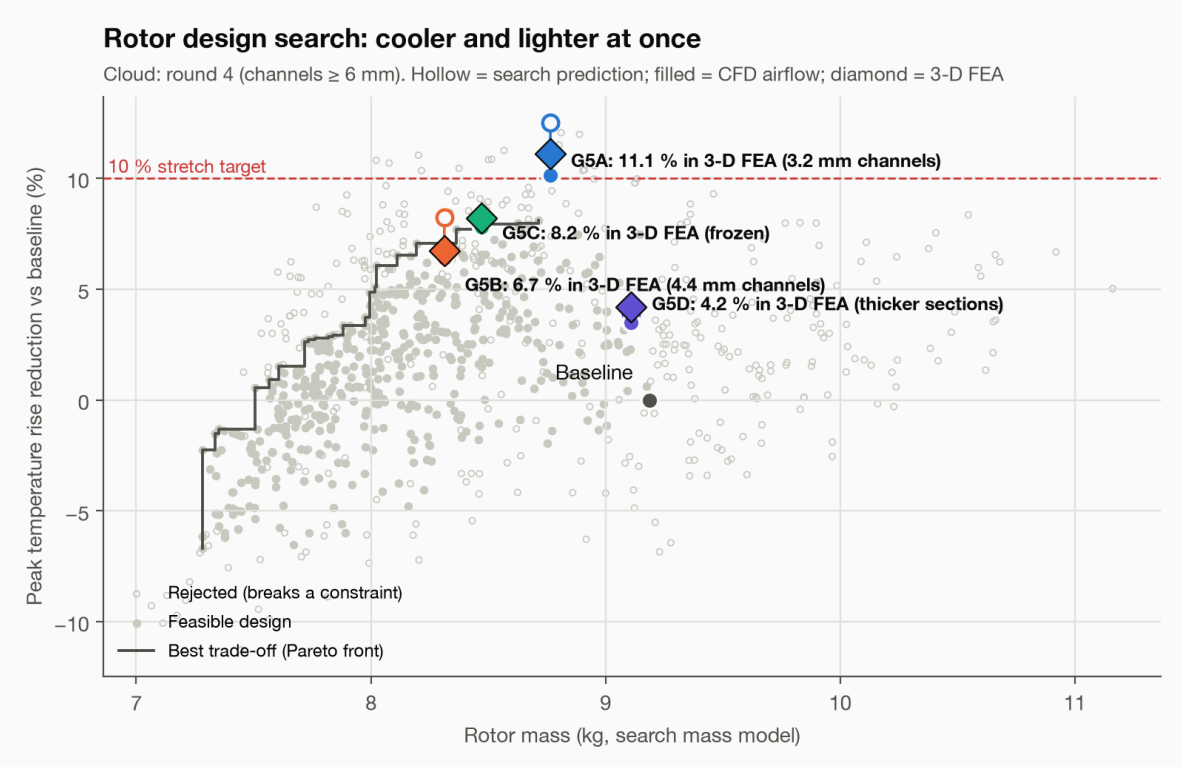


Figure 18: Round 4 search (channels ≥ 6 mm): all designs, Pareto front and verified candidates

	C0	G5A	G5B	G5C	G5D
ϕ predicted (CFD mesh scale)	n/a	0.185	0.204	0.409	0.340
ϕ from CFD	0.235	0.150	0.171	0.403	0.321
Vent air mass flow, 80 km/h	16.1 g/s	13.5 g/s	15.8 g/s	36.0 g/s	25.4 g/s
LC-04 peak, two-zone model with CFD ϕ	592.8 °C	535.2 °C	556.5 °C	548.5 °C	572.9 °C
LC-04 peak, 3-D FEA	**592.4 °C**	**529.4 °C**	**554.2 °C**	**545.9 °C**	**568.5 °C**
Rise change, 3-D	reference	−11.1 %	−6.7 %	−8.2 %	−4.2 %

ASSESSMENT. The ϕ regression over-predicted G5A and G5B by about 20 % (edge of the data) and was within 2–6 % for G5C and G5D (inside the data). The two-zone model agreed with 3-D FEA within 6 °C on all five geometries, none of which it was calibrated on. No search result was used without its own CFD and 3-D run.

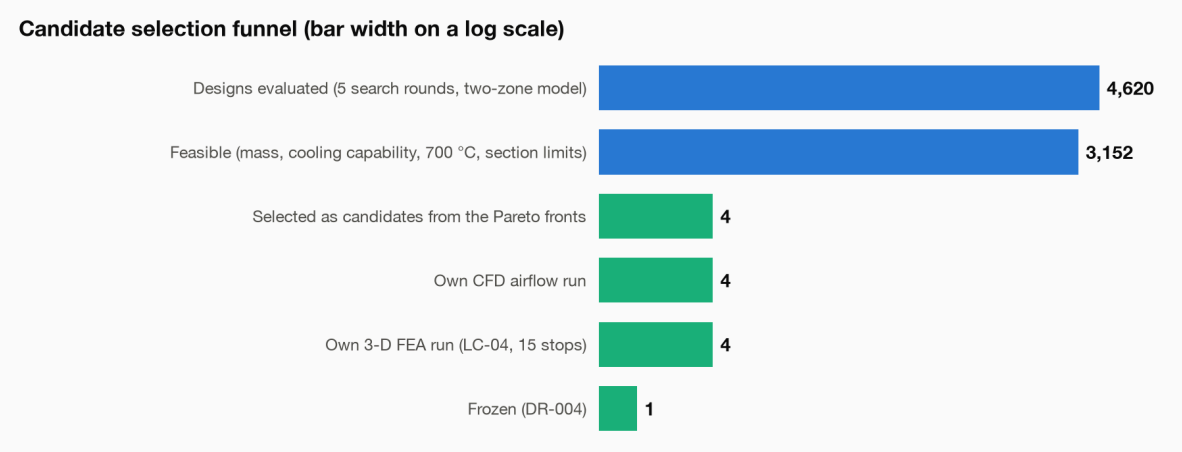


Figure 19: Candidate selection funnel



Figure 20: ϕ regression against CFD for the DoE (left out of the fit) and the candidates

11.5 The frozen design, G5C

G5C keeps the outside envelope and hat. It has 62 straight radial vanes of 3.3 mm, friction plates of 7.1 mm (vent gap 13.8 mm) and vanes from Ø186.5 mm. It moves 2.2 times the baseline's air through the vents. Straight radial vanes are non-directional: one part fits both sides of the vehicle.

Stop by stop. G5C is not cooler from the first stop. Its thinner plates hold less heat under the pad: it runs 7–15 °C hotter than the baseline for stops 1–6, equal at stop 6, and cooler from stop 7, ending 46 °C cooler after 15 stops.

Requirements (two-zone model with CFD airflow for LC-01/02/03/05/07; 3-D FEA for LC-04; CAD for geometry):

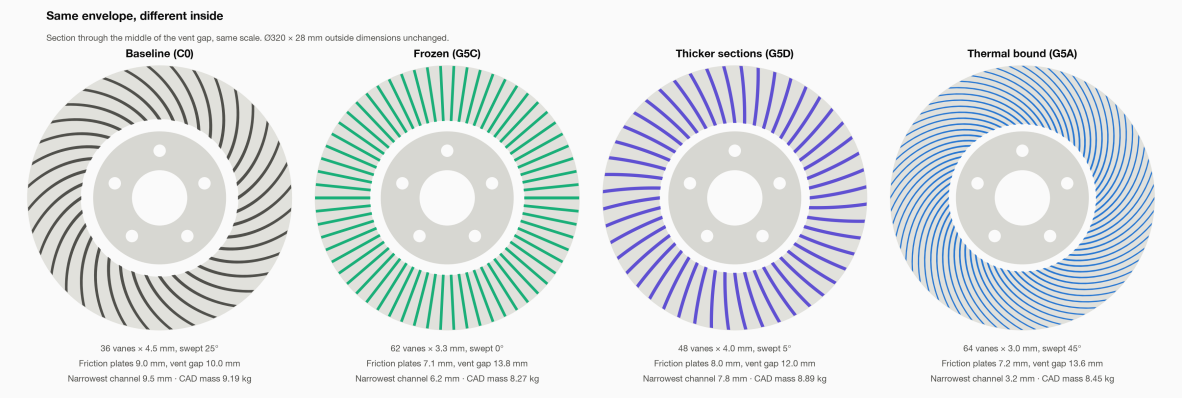


Figure 21: Vane layouts at mid-gap, same scale: C0, G5C, G5D, G5A

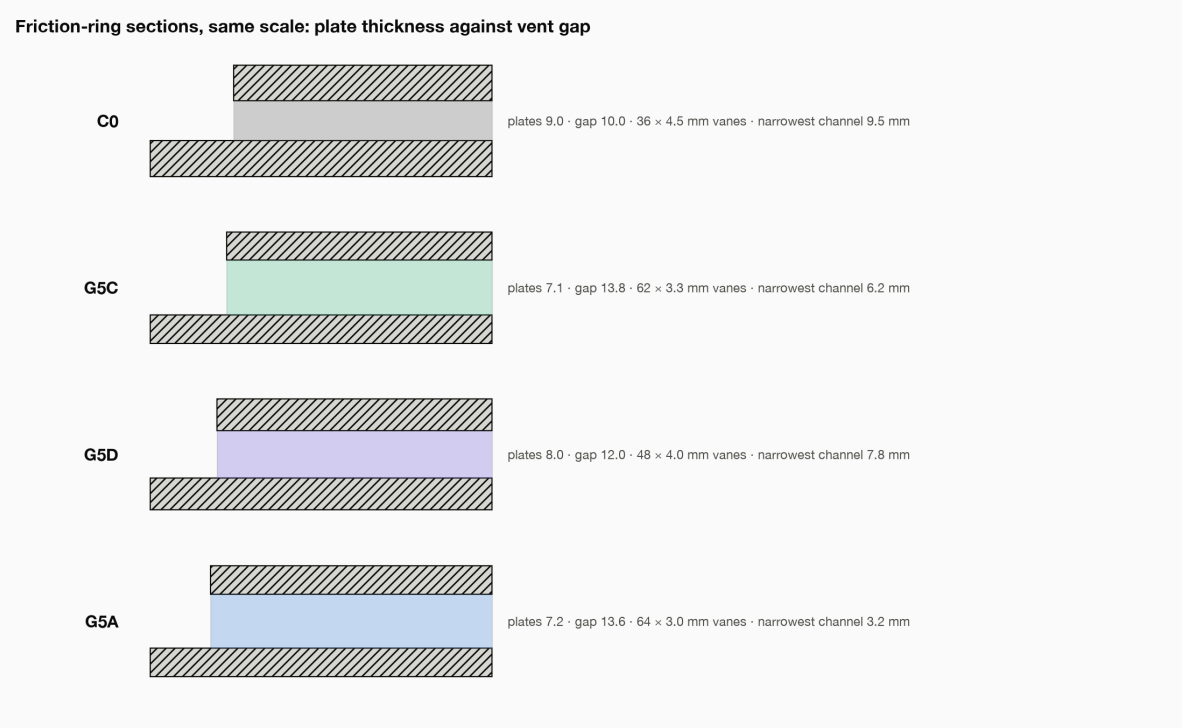


Figure 22: Friction-ring sections, same scale: plates, vent gap and narrowest channel

Requirement	Limit	C0	G5C (frozen)	G5D	G5A
REQ-THM-001, LC-04 rise (3-D)	≥ 10 % lower (stretch)	reference	−8.2 %: not met	−4.2 %: not met	−11.1 %: met
REQ-THM-002, WLTC / NEDC / WLTC×3 peak	≤ baseline	171 / 138 / 194 °C	161 / 137 / 170 °C: pass (NEDC margin 0.4 °C)	166 / 138 / 182 °C: NEDC +0.1 °C, marginal fail	156 / 136 / 164 °C: pass
REQ-THM-003, LC-04 (3-D) / LC-05 / LC-07 peak	≤ 700 °C	592 / 642 / 630 °C	546 / 622 / 630 °C: pass	569 / 630 / 627 °C: pass	529 / 617 / 622 °C: pass

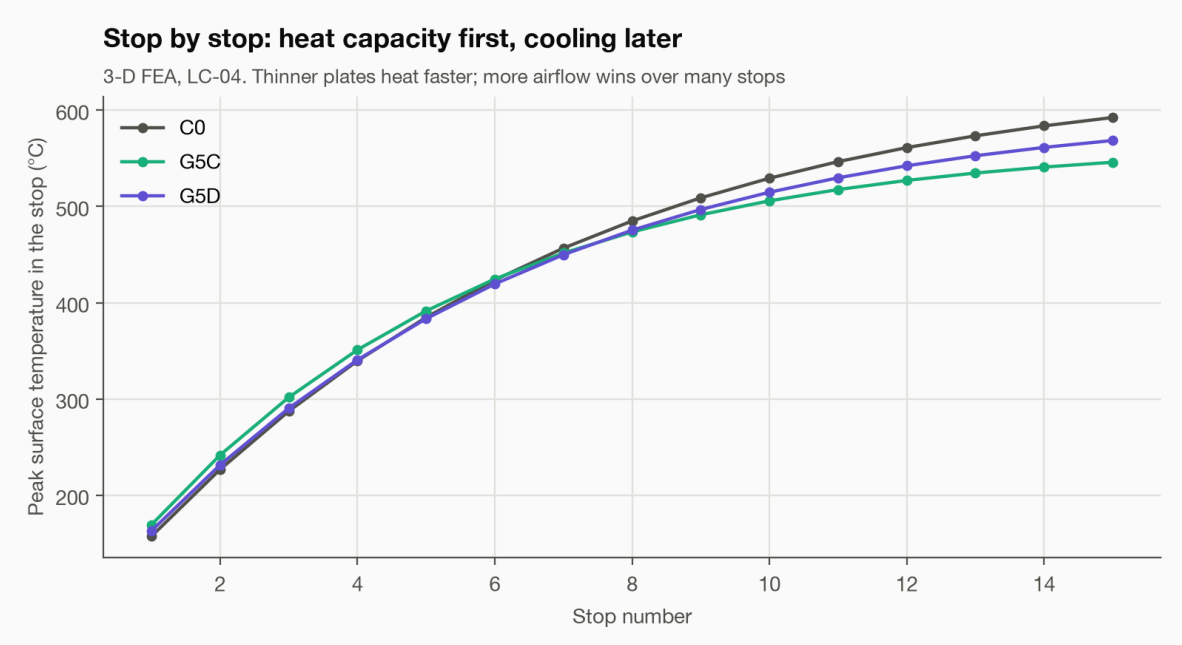


Figure 23: LC-04 peak surface temperature per stop, 3-D FEA: C0, G5C, G5D

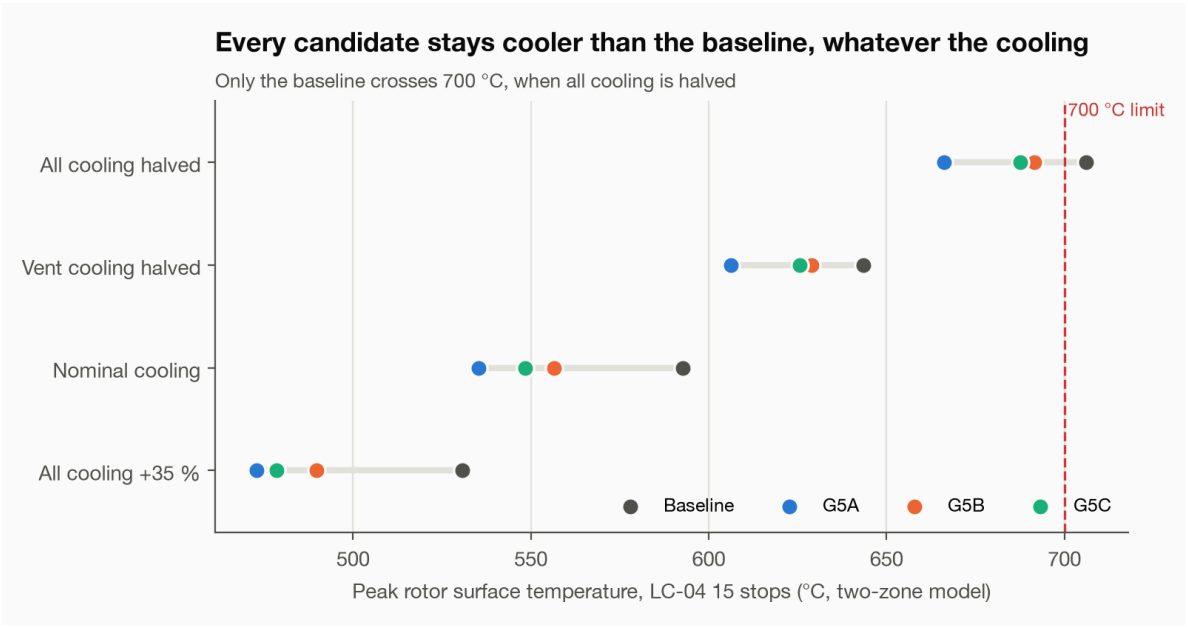


Figure 24: LC-04 peak temperature under changed cooling assumptions, two-zone model

Requirement	Limit	C0	G5C (frozen)	G5D	G5A
REQ-THM-004, cooling capability at 100 km/h	≥ baseline	7.7 W/K	10.2 W/K: pass	8.7 W/K: pass	10.8 W/K: pass
REQ-GEO-001, envelope	Ø320 × 28 mm, same hat	n/a	pass	pass	pass
REQ-GEO-002, mass (CAD)	≤ 9.19 kg	9.19 kg	8.27 kg: pass	8.89 kg: pass	8.45 kg: pass
Robustness: LC-04 with all cooling halved	≤ 700 °C	706 °C	688 °C: pass	694 °C: pass, small margin	666 °C: pass
Narrowest vent channel (DFM, assumed limit)	≥ 6 mm	9.5 mm	6.2 mm: pass, unconfirmed	7.8 mm: pass, unconfirmed	3.2 mm: fail

Robustness (two-zone model, LC-04): G5C stays cooler than the baseline when all cooling is scaled by 1.35, when the vent cooling is halved, and when all cooling is halved; the gain ranges from –10.3 % to –2.7 %. With all cooling halved the baseline reaches 706 °C and G5C 688 °C.

DECISION D-04 (G5, 2026-11-01, owner, DR-004). G5C frozen as the G5 design. G5A recorded as the thermal upper bound of the design space. Open: manufacturability review, G6 checks.

11.6 Splitter vanes (post-freeze check)

OPT-001 Sec. 11. Splitter vanes cannot be added to G5C as it is: the split channels at the rim would be about 4.8 mm. Layouts with fewer full vanes plus splitters (31–48 + 31–48) were screened: total vane surface 0.96–1.04 times G5C, CFD airflow –5 % to +1 %, LC-04 rise within ±1 % of G5C.

ASSESSMENT. No measurable benefit. With a minimum channel width, splitters redistribute vane surface but do not add it, and they did not raise the airflow. Layout S31 (31 + 31) matches G5C to about 1 % with a 15.6 mm inlet and is recorded as a manufacturing fallback.

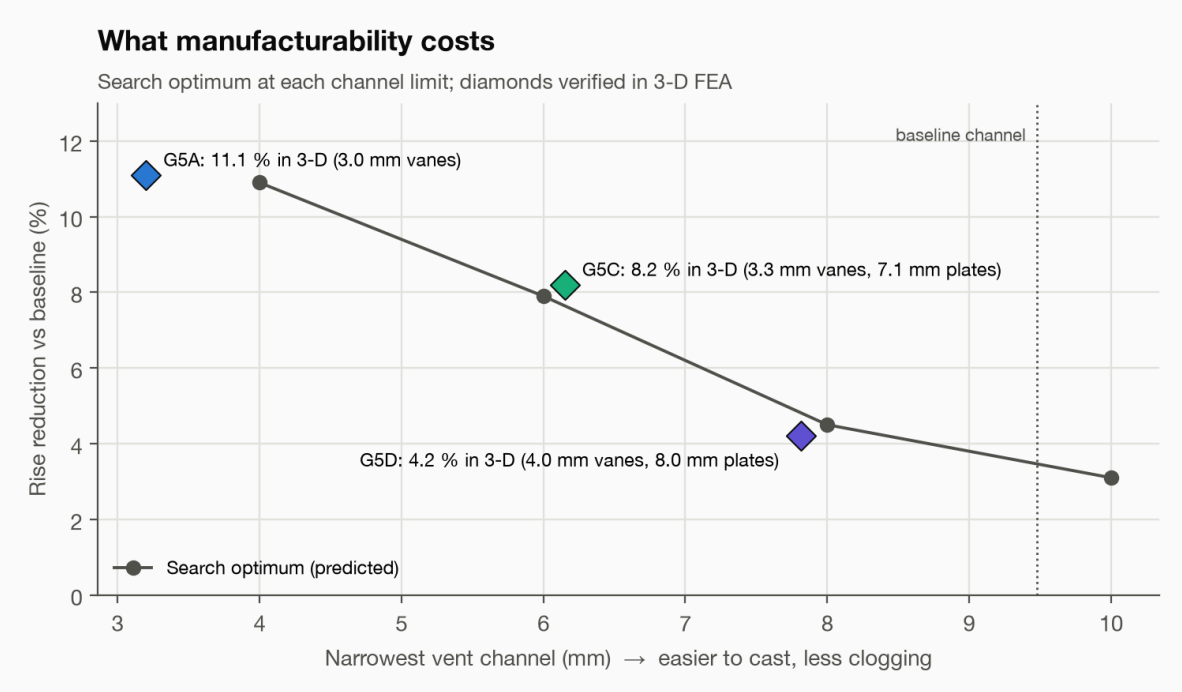


Figure 25: Temperature-rise reduction against the narrowest vent channel: search optima and 3-D verified designs

12. Manufacturability and the practical design

12.1 Why this matters

The vent channels of a cast-iron rotor are formed by a resin-bonded sand core. The narrowest channel is the thinnest section of that core; thin cores break and are hard to compact, and narrow channels collect brake dust and road dirt in service. Thin metal sections (vanes, plates) cool fast in the mould, which risks incomplete filling and hard, brittle spots, and thin plates reduce the wear allowance. None of this is in the thermal model.

12.2 What was found

No published minimum vent channel or vane thickness for grey-iron brake discs was found; foundries set these limits. General sand-casting guidance gives minimum metal sections of about 3–5 mm. The baseline has 9.5–14.5 mm channels, 4.5 mm vanes and 9 mm plates. The 6 mm channel limit of round 4 is an assumption.

12.3 Trade-off

The search was repeated with different limits. A further run with thicker sections (channels ≥ 6 mm, vanes ≥ 4 mm, plates ≥ 8 mm, all assumed) produced G5D, which was verified with CFD and 3-D FEA.

Option	Vanes	Narrowest channel	Plates	LC-04 rise change	Basis
G5A	64 × 3.0 mm, swept 45°	3.2 mm	7.2 mm	–11.1 %	3-D FEA
Search, channel ≥ 4 mm	64 × 3.2 mm, swept 30°	4 mm	7.1 mm	–10.9 %	prediction
G5C	62 × 3.3 mm, straight	6.2 mm	7.1 mm	–8.2 %	3-D FEA
G5D	48 × 4.0 mm, swept 5°	7.8 mm	8.0 mm	–4.2 %	3-D FEA
Search, channel ≥ 8 mm	50 × 3.1 mm, straight	8 mm	7.6 mm	–4.5 %	prediction
Search, channel ≥ 10 mm	39 × 3.1 mm, straight	10 mm	7.7 mm	–3.1 %	prediction

12.4 G5D against G5C

Item	G5C	G5D
LC-04 rise change, 3-D	–8.2 %	–4.2 %
Mass (CAD)	8.27 kg (–0.92 kg)	8.89 kg (–0.30 kg)
Vane thickness / plate thickness	3.3 / 7.1 mm	4.0 / 8.0 mm

Item	G5C	G5D
Narrowest channel	6.2 mm	7.8 mm
Single 1 g stop (LC-06) vs baseline	+5 °C	+2 °C
LC-04 stops 1–6 vs baseline (3-D)	up to +15 °C; cooler from stop 7	up to +6 °C; cooler from stop 5
WLTC / NEDC / WLTC×3 vs baseline	−11 / −0.4 / −24 °C	−6 / +0.1 / −12 °C
All cooling halved	688 °C	694 °C
Vent air mass flow	36.0 g/s	25.4 g/s
Same part for both sides	yes (straight)	no (5° sweep; could be set to 0° with little change)

G5D's NEDC peak is 0.1 °C above the baseline. This fails REQ-THM-002 as written but is far inside the model uncertainty; a 3-D NEDC run would be needed to resolve it.

ASSESSMENT. The thicker-section design keeps about half of G5C's temperature reduction and a third of its mass saving, with 27 % wider channels, 21 % thicker vanes and 13 % thicker plates. Without foundry data it is the lower-risk design to make; G5C remains the better design if its sections are confirmed castable. The choice depends on a foundry limit that this project does not have.

DECISION D-05 (open, owner, DR-005). Production-intent variant: (a) keep G5C, subject to foundry confirmation of 3.3 mm vanes, 7.1 mm plates and 6 mm channels; or (b) adopt G5D as the manufacturing-robust variant, accepting about half the thermal gain. Recommendation: hold G5C as frozen for G6, carry G5D as the declared fallback, and decide at the foundry review.

13. What the result does and does not show

G5C is optimal for the stated objective, with these models and constraints. It does not show that straight radial vanes are better than curved vanes in general, or that G5C is the rotor a production programme should build.

Reason straight vanes won here	Real or model artefact
Straight radial vanes pump more air than backward-curved ones at the same speed (fan theory; CFD agrees)	Real
The curved vanes studied bend at a constant rate with no designed inlet angle; production curved vanes are shaped to the incoming flow	Model artefact (favours straight)
The vent heat-transfer correlation treats channels as straight; curved channels produce secondary flows that raise heat transfer	Model artefact (favours straight)
CFD heat-transfer check not in the model; if it is right, vent-driven gains shrink, G5C's most	Open
Rotor in still air; no wheel, caliper or forward speed	Simplification

Criteria a production programme weighs and this study did not optimise: noise (vane count and pattern set the disc's vibration modes), thermal cracking and distortion, pumping loss (fuel and energy consumption), cost and castability, wear allowance, brake feel, corrosion. In industry, straight radial vanes are common in mass-market rotors (non-directional) and curved vanes in many performance rotors; pillar layouts are used for noise and cracking.

ASSESSMENT. This study answers one criterion with quantified uncertainty. Selecting a production rotor needs analysis of the other criteria and engineering judgement. A fairer straight-versus-curved comparison needs curved vanes with a designed inlet angle and a heat-transfer model for curved channels.

14. AI-assisted working method: assessment

LOG.md records 22 entries to G5 (2 of them gate decisions). The others are cases where AI-produced work was wrong or incomplete, or where the owner's review changed the work. By type:

Type	Examples	How caught
Wrong number or model error	Lumped vane area 25 % high; 3.5 % heat-input loss with a first explanation that was wrong; load-schedule bug	Independent cross-checks (mass, energy), regression tests
Not converged, presented as converged	First CFD mesh series; first CFD heat run	Further refinement; physical plausibility
Wrong source reading	Equations read from a corrupted PDF text layer; secondary web summaries quoted as primary	Page images; book's worked example; primary documents
Optimiser exploited model gaps	Three rounds (Section 11.3); castability found only at verification	Own CFD and 3-D run per candidate
Scope not stated	Optimum reported without its limits; thin sections not considered	Owner's questions (Sections 12, 13)
Presentation	Captions not matching data; chart axis clipping a data point; overlapping text	Inspection of every output

ASSESSMENT. The assistant shortened implementation and iteration substantially. The errors that mattered were not syntax errors but plausible-looking results: an unconverged number, a model gap, an unstated scope. Each was caught by an independent check or by the owner's question, not by the assistant's own confidence. Verification and the judgement of what to trust remained with the engineer.

15. Conclusions and recommendation

Objective (restated). Lower the LC-04 peak surface temperature of the front rotor without added mass, inside the original envelope.

Conclusions.

1. **The vent airflow is the lever.** With the envelope and friction faces fixed, the internal geometry acts mainly through the air pumped through the vents (Section 2). CFD showed that the baseline's swept vanes move 16 g/s at 80 km/h; straight radial vanes with a taller vent gap move up to 36 g/s (Sections 9, 11.2).
2. **Airflow and heat capacity pull against each other.** Thinner plates and more vanes cool the ring faster between stops but let each stop heat it more. Every optimised design is hotter than the baseline in the first stops of LC-04 and cooler later; the benefit exists only under repeated braking (Section 11.5).
3. **Castability bounds the result.** The verified gain falls as the minimum vent channel widens: –11.1 % at 3.2 mm, –8.2 % at 6.2 mm, –4.2 % at 7.8 mm (Section 12.3). Where a production design sits on that curve is set by foundry capability, not by simulation.
4. **The optimum is specific to this objective and these models** (Section 13). Noise, cracking, pumping loss and cost were not optimised, and the CFD heat-transfer discrepancy is open.

Comparison of the final designs (3-D FEA, LC-04 15 stops):

Design	Peak, °C	Rise, %	Mass, kg	Gain mechanism	Main penalty or risk	Status
C0	592.4	ref.	9.19	Baseline	n/a	Reference
G5C	**545.9**	**–8.2**	**8.27**	62 straight radial vanes, taller vent gap: 2.2 × the vent air	Sections not confirmed castable (3.3 mm vanes, 7.1 mm plates, 6.2 mm channels); up to +15 °C in stops 1–6	**Recommended, frozen (D-04)**
G5D	568.5	–4.2	8.89	48 near-straight 4 mm vanes: 1.6 × the vent air	Half of G5C's gain; NEDC +0.1 °C (inside model uncertainty)	Manufacturing fallback (D-05)

Design	Peak, °C	Rise, %	Mass, kg	Gain mechanism	Main penalty or risk	Status
G5A	529.4	–11.1	8.45	64 thin swept vanes	3.2 mm channels, judged not castable	Thermal upper bound only

Why G5C. It gives the largest verified reduction among the designs that meet the assumed minimum channel width (6 mm). It meets every G5 requirement except the 10 % stretch target, and it is the lightest design. Its straight vanes also make it non-directional, so one part fits both sides of the car. It achieves this by trading plate thickness, and so heat capacity, for vent airflow. The trade-off it accepts is a hotter rotor in the first few stops and thinner sections than the baseline.

ASSESSMENT (final). For the stated objective, the best design found is **G5C**: –8.2 % temperature rise and 0.92 kg lighter than the baseline. If the foundry review does not confirm its 3.3 mm vanes and 6 mm channels, **G5D** is the fallback, at about half the gain. Neither design is a production release. The G6 checks (wear limit, remaining load cases in 3-D, thermal stress) and the criteria in Section 13 still apply, and they need engineering judgement beyond this study.

16. Open items and next steps (G6)

Item	Action	Gate
Wear limit (REQ-GEO-003)	LC-04 at 26 mm thickness for G5C (and G5D)	G6
LC-05 and LC-07 in 3-D	Confirm the two-zone results	G6
NEDC margin	3-D NEDC run if a margin is required (G5C +0.4 °C margin, G5D –0.1 °C)	G6
Thermal stress, coning (REQ-STR-001)	Thermo-mechanical run, report only	G6 stretch
Manufacturability (D-05)	Foundry review of minimum vane, plate and channel sections	external
CFD heat transfer	Wall-resolved thermal CFD (prism layers, $y+ \approx 1$) or test data	future
Noise	Modal analysis of G5C (62 vanes)	future
Traceability (REQ-DOC-001)	DVPR closure	G6

Appendix A. Document register

No.	Document	Content
01	PRG-001	Programme charter
02	PRG-002	Gate plan
03	PRG-003	Tool qualification
04	BMK-001	Benchmark study
05	VTS-001	Vehicle technical specification
06	DR-001	Reference vehicle and baseline rotor
07	SRS-001	System requirements
08	LC-001	Load cases
09	DVPR-001	Verification plan and report
10	REF-001	References
11	CAD-001	Baseline rotor (parametric CAD)
12	TM-001	Lumped thermal model
13	CON-001	Concept comparison
14	DR-002	Lead concept
15	CAE-001	3-D thermal FEA

No.	Document	Content
16	VV-001	Uncertainty budget
17	DR-003	Airflow CFD scope
18	CFD-001	Vent airflow CFD
19	TM-002	Two-zone thermal model
20	CFD-002	Airflow design of experiments
21	OPT-001	Design optimisation
22	DR-004	Design freeze
23	DR-005	Production-intent variant (open)

Appendix B. Principal assumptions

Item	Value	Status
Rotor internal geometry (vanes, plates, vane ID)	Baseline values in Section 6	Representative, assumed
Material	Grey iron, temperature-dependent k, cp; ρ 7,150 kg/m³	Secondary sources
Heat partition to disc	90 %	Assumed
Emissivity	0.6	Assumed
Front brake bias	70 %	Assumed
Test mass / GVM	1,850 / 2,150 kg	Assumed / one public source
Pad track	r = 100–150 mm	Assumed
Vent heat transfer	Limpert vane correlation with effectiveness cap	Literature; not confirmed by CFD
Vent airflow	CFD, still air, 80 km/h; regression for other geometries	Simulation
Temperature limit	700 °C	Literature-based project limit
Minimum channel / vane / plate (DFM)	6 / 4 / 8 mm	Assumed; no foundry data